

Sail Bhandekar

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Portfolio:- [Portfolio](#)

Embedded C Developer with **1.5+ years of experience** in **IoT embedded systems development** and **firmware design**. Skilled in **C, Embedded C programming, RTOS-based application development, and low-level driver development on microcontrollers**. Strong expertise in **real-time debugging, peripheral interfacing, and communication protocols**. Seeking to contribute to innovative engineering teams as an **Embedded Software Developer** or **Embedded Engineer**, delivering **reliable and efficient embedded software solutions** for **real-time and resource-constrained systems**.

Skills:

Programming Language	C, Embedded C, C++
Tools & Technologies	Keil µVision, STM32CubeIDE, Simplicity Studio 5, JTAG debugging, Proteus simulation, GCC & Makefile, RTOS-based firmware development, low-level driver development, CAN-based systems, Raspberry Pi IoT gateway, MQTT communication
Protocols	CAN, SPI, UART, I2C, BLE, MQTT
Standards & Guidelines	CAN Protocol (ISO 11898), MISRA C, RTOS system design principles.
Testing and debugging	Functional Testing, System-Level testing, real-time issue analysis
Operating Systems	RTOS (real-time scheduling & multitasking), Linux, Windows

Professional Experience:

Software Engineer – Votary Softech Solutions Private Limited

Hyderabad, India | Jun 2025 – Present

- Developed **RTOS-based firmware** in **Embedded C** for **Smartwatch devices** using **Silicon Labs EFR32MG24 microcontrollers**, implementing **real-time task scheduling** for system responsiveness.
- Integrated motion and health sensors (accelerometer, heart-rate)** via **SPI/I2C** and **developed low-level drivers** for **sensor data acquisition and processing**.
- Implemented low-power management techniques** to **optimize battery life** and **designed SPI-based display drivers** for **real-time UI rendering**.
- Engineered a **factory automation system** using **embedded sensor nodes** and a **Raspberry Pi 4 IoT Gateway**, enabling **data aggregation and cloud connectivity via Bluetooth Mesh**.

Electronic Engineer – Win Render Systems LLP

Pune India | Aug 2023 – Aug 2024

- Developed and optimized **embedded C firmware** for industrial control systems using **LPC2148 microcontrollers**.
- Implemented **real-time control logic** for a **Tablet Metal Detector system** to ensure accurate detection and system monitoring.
- Designed **USART-based communication** to transfer operational data and system status to an **HMI display** for user interaction.
- Automated **peripheral testing routines** to improve testing efficiency and system stability.
- Enhanced firmware features based on **customer requirements**, including performance optimization and display updates.
- Utilized **Keil µVision and Proteus** for firmware development and simulation, along with **JTAG** for on-device debugging.

Internship

Embedded System Engineer Intern – Hans Cag Electronics Pvt. Ltd

Remote | Sept 2024 – May 2025

- Gained foundational knowledge of **microcontroller architecture, C programming, and hardware-software integration**.
- Developed prototype-level embedded applications** on **ESP32**, interfacing **sensors and modules** using **GPIO, UART, I2C, and PWM**.
- Wrote, debugged, and tested firmware** using **Embedded C** in the **Arduino IDE** and **performed system simulation** in **Proteus**.

Project Highlights:

Project Title: Smartwatch Firmware Development.

Platform	Microcontroller – Silicon Labs EFR32MG24 (xG24 Series)
Customers Handled	Silicon Labs
Programming Language	Embedded C.
Tools Used	Simplicity Studio 5, JTAG Debugger
Description	Developed RTOS-based firmware for a smartwatch, featuring real-time task scheduling, sensor integration (accelerometer, heart-rate), and low-power optimization. Designed SPI/UART/I2C drivers and an SPI display driver for real-time data visualization.

Project Title: Factory Automation System Using Embedded Controllers and IoT Gateway.

Platform	Silicon Labs EFR32MG24 (xG24 Series) Sensor Nodes & Raspberry Pi 4 IoT Gateway
Customers Handled	Industrial Automation Environment (Internal Project: - Votary tach)
Programming Language	Embedded C,
Tools Used	Simplicity Studio 5, JTAG Debugger,
Communication Technologies	Bluetooth, Bluetooth Mesh
Description	Built a scalable factory automation system using embedded controller nodes and a Raspberry Pi IoT Gateway. Implemented Bluetooth Mesh networking for device communication, enabling real-time process monitoring, data aggregation, and cloud connectivity.

Project Title: Tablet Metal Detector Control System .

Platform	LPC2148 (ARM7)
Customers Handled	Internal Project: - Win Render Systems LLP
Programming Language	Embedded C.
Tools Used	Keil µVision, Proteus Simulation, JTAG Debugger
Description	Tablet Metal Detector system using LPC2148 microcontrollers. Implemented real-time control logic and designed USART-based communication to interface with an HMI display for system monitoring and user interaction, automated peripheral testing, and enhanced firmware features based on customer requirements to improve performance and reliability.

Certifications & Courses

- Advance Embedded Systems Course – Vector India (Sept 2024 –Jun 2025)
 - Includes Linux kernel programming, and system software fundamentals,.
- Qualcomm_AI Upskilling Certificated [Link](#)

Scholastics:

Degree	Institute/College/School	University/Board	Year	%
Bachelor of Engineering	JSPM Narhe Technical Campus	SPPU, PUNE	2024	7.4 CGPA
Diploma	Government Polytechnic, Bramhapuri	MSBTE	2021	83.65%
10th STD	Navbharat Vidyalaya,Vyahad Buj	State Board, Maharashtra	2018	80.20 %